

Table 1: Comparison of the features of three generations of database systems

Feature / Data management types	DBMS	NoSQL	NewSQL
Relation between entities	✓	✗	✓
SQL support	✓	✗	✓
ACID transaction handling	✓	✗	✓
Horizontal scalability	✗	✓	✓
High performance while handling big volumes	✗	✓	✓
Schema-less structure	✗	✓	✗

relation between entity models, ACID transaction handling capability, etc. Popular NewSQL database systems are VoltDB, ClusterixDB, CockroachDB and NeoDB.

Polyglot databases in cloud platforms

A data store is an integral component of any application development as it is the persistent store for handling all the data required for processing logic. Database solutions are moving towards polyglot databases, which combine multiple types of databases for an enterprise architecture. Polyglot databases are typically used in microservices architecture as they use multi-types of database systems in an integrated solution.

Such an integrated architecture is needed to ensure high performance for any type of data processing across different services. Typically used in cloud adoption, this kind of implementation is supported by DBaaS (database as a service). Data-driven thinking helps build enterprise architecture with a flexible data design.

DBaaS is a cloud based database management system (e.g., Aurora DB). It is highly scalable and offers enhanced security in cloud architecture. Suitable for large enterprise design, it offers shared infrastructure with

the availability of monitoring and performance tuning tools.

Columnar databases

A database system is a matrix representation of rows (records) and columns (fields) of data accessed through queries. In a traditional RDBMS, a standard SQL query scans through each record, filters it based on the query condition, and populates the data in the required result set. This works well for small and medium applications. But when it comes to large solutions like data warehouse applications, where the number of columns and rows are far more, there are huge performance drops due to physical I/O during query operations when scanning through records and filters for required results.

NoSQL databases, which are schema less and don't store in rows and columns but in key-value, document, graph or column-oriented data models, address this situation better with semi-structured or unstructured database models.

A columnar database is completely different from a traditional data structure as it stores data in columns and not in rows, and helps in querying for data with less disk access. This helps reduce the physical I/O and optimises the workload.

This database is suitable for large data processing systems like data warehouses, business intelligence systems or retail store operations. As it is self-indexed, managing data indexing for faster data access is easy.

Some examples of columnar databases that are very popular are MariaDB, ClickHouse, Apache Parquet, MonetDB and Apache HBase.

CAP theorem in cloud data architecture

Any cloud architecture is built on certain key expectations, which include high-availability, cost-efficiency, security compliance and agility in design. The CAP theorem was conceived by Brewer (and hence is also called the Brewer theorem) in 1998 and then formally in 2000 to address the non-functional and architectural

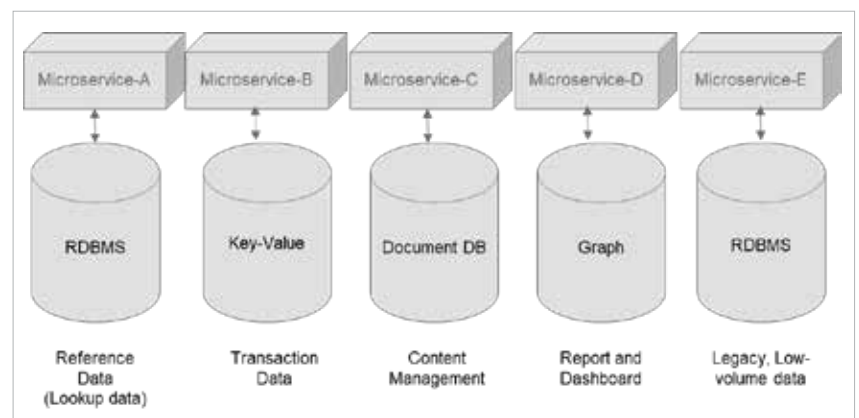


Figure 1: Polyglot data architecture